Exponential Applications 2

1. **Annuity**

An **annuity** is a financial contract that provides a series of regular, fixed payments over a specific period.

Depending on your goal, an annuity functions in one of two ways:

- **Accumulation Phase (Building):** You make regular contributions to an account that grows with interest over time to reach a future goal.
- **Distribution Phase (Receiving):** You start with a lump sum that is paid out to you in periodic installments until the balance reaches zero.

1.1. Building An Annuity - Future Value

The Formula

$$FV = P \times \left[\frac{(1+r)^n - 1}{r}\right]$$

Where:

- FV: The Future Value (the total amount you want to have).
- P: The Periodic payment (the amount you deposit each period).
- r: The interest rate per period (expressed as a decimal).
- n: The total number of periods (e.g., total years or months).

Example: A rookie player in the NFL just signed his first 7-year contract. To prepare for his future, he deposits $150,000 at the end of each year for 7 years in an account with 3.1% compounded annually.

a. How much will he have in his account after 7 years?

b. How much interest was earned over the 7 years?

You try: A new professional basketball player signs a 6-year contract. To save for the future, he deposits $120,000 at the end of each year into an investment account that earns 3.4% interest compounded annually.

a. How much money will be in the account after 6 years?

b. How much interest was earned during the 6 years?

1.2. Receiving An Annuity- Present Value

The Formula

$$PV = P \times \left[\frac{1-(1+r)^{-n}}{r}\right]$$

Where:

- PV: The Present Value (the lump sum you start with today).
- P: The Periodic payment (the amount you receive each period).
- r: The interest rate per period (as a decimal).
- n: The total number of periods.

Example : A wealthy family member has funded an annuity that will pay you $1500 at the end of each year for six years. If the interest rate is 8%, compounded annually, find the present value of the annuity.

You try: A relative has set up an annuity that will pay you $2,000 at the end of each year for 5 years. The account earns 7% interest compounded annually. Find the present value of the annuity today.

Example: To supplement his pension in the early years of his retirement, Ryan plans to use $124,500 of his savings as an ordinary annuity that will make monthly payments to him for 20 years. If the interest rate is 2.8%, how much will each payment be?

You try: To fund his extensive pickleball tournament travel and new paddle upgrades for the next 15 years, Sarah decides to invest $\$150,000$ into an ordinary annuity. If the annuity offers an annual interest rate of 3.5% compounded monthly, how much will Sarah receive in monthly payments?

2. **Loans and Amortization**

Amortization is the process of paying off a debt over a set period through a series of regular, fixed payments.

$$M = L \times \left[\frac{r(1+r)^n}{(1+r)^n - 1} \right]$$

Variable Guide

- M: Monthly Payment (The amount you pay each period).
- L: Loan Amount (The total principal you are borrowing today).
- r: Monthly Interest Rate (Annual Rate / 12).
- n: Total Number of Payments (Loan Term in years $\times$ 12).

Example: The average used car price is $32,000. The average interest rate is currently 6.39% for a 60-month loan (5 years). If you have $5,000 for a down payment, what can you expect the monthly payment to be? How much will you pay in total for the car?

You try: The average rate for a 30-year fixed mortgage is 7.5%. Assume a down payment of 20% on a home purchase of $315,000.

a. Find the monthly payments needed.

b. How much will be paid in total over the life of the loan?

You try: In 2023, the average amount of student loan debt for a bachelor's degree recipient at a private university was $54,921. Assume an interest rate of 6.87%

a. Find the monthly payment and calculate the total interest paid over the course of the loan for a 10-year payment plan.

b. Find the monthly payment and calculate the total interest paid over the course of the loan for a 20-year payment plan.